The Price Effects of Prohibiting Price Parity Clauses: Evidence from Global Hotel Chains

Prohibiting Price Parity Clauses

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Abstract: Dominant platforms like Booking.com have often imposed Price Parity Clauses to prevent lower prices on alternative sales channels. We provide quasi-experimental evidence on the removal of these price restrictions in France in 2015 for three major global hotel groups. Our analysis reveals limited and non-significant price effects for rooms sold through consumer-visible channels, such as hotels' websites or online travel agencies. However, we document a significant price reduction on sales channels not visible to consumers, such as direct offline bookings. Additionally, we identify a significant shift in the share of bookings from online travel agencies to the hotels' direct offline channels.

Keywords: price parity clauses, online travel agents, platform regulation.

Classification: D40, K21, L10, L42, L81

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1 Introduction

In today's highly digitised economy, goods and services can be purchased directly from sellers or through intermediary platforms. In online markets, the contractual relationship between the involved parties often follows the agency model, whereby sellers decide the final prices displayed on each sales channel, including those offered by the platforms. For every intermediated transaction, the platforms receive a commission fee, which is usually proportional to the transaction price. For example, hotels can offer rooms on their own website or through Online Travel Agencies (OTAs) such as Booking.com or Expedia. If a room is reserved through an OTA, the hotel will pay a commission fee to that OTA. It is, therefore, in the platforms' best interest to maximise the number of transactions that consumers finalise through them, and for this purpose, they may adopt specific contractual arrangements.

Controversial arrangements at the centre of regulatory scrutiny are Price Parity Clauses (PPCs), namely, price restrictions imposed by platforms on client sellers. These clauses stipulate that the latter cannot charge lower prices on alternative sales channels. PPCs are widespread in the e-commerce and lodging sectors, but also exist in industries such as entertainment, insurance, and payment systems. The so-called "wide" PPCs mandate that the price charged by sellers cannot be lowered on *any* alternative sales channel. "Narrow" PPCs are seemingly less rigid, as they allow sellers to lower prices on rival platforms and offline but not through their website.

Platforms affirm that PPCs are necessary to prevent showrooming, where consumers initially browse the platform to identify their preferred seller but then switch to the seller's direct channel to obtain a discount. This practice, if widely adopted by consumers, could render platforms' activity unprofitable, possibly undermining their existence. PPCs make free-riding unlikely as consumers cannot find lower prices elsewhere. Another important argument presented by platforms in defence of PPCs is that showrooming may undermine their incentives to invest in improving the quality of the services provided to both sellers and consumers.

On the other hand, competition authorities and regulators claim that PPCs reinforce the dominant position of leading platforms and contribute to maintaining higher prices for consumers. Indeed, if sellers cannot differentiate prices, consumers are more likely to make purchases through platforms, which generally offer additional benefits. Platforms can then impose relatively high commission rates and extract a large portion of the sellers' profits. Conversely, if PPCs were removed, sellers could lower prices when selling directly or on rival platforms, thereby limiting the ability of dominant platforms to charge excessive commissions.

This article focuses on the lodging sector and investigates the effects of prohibiting all types of PPCs. We exploit the first-of-its-kind policy change that occurred in France, the Macron Law, which was adopted on 9 July 2015 and promulgated on 6 August 2015. Our unique and comprehensive dataset covers three years, from June 2014 to May 2017, and includes monthly transaction data for 166 hotels affiliated with three of the top 10 hotel groups operating in Europe and worldwide. These hotels span 61 cities in seven European countries and employ multiple channels to sell their rooms. Among these channels are major OTAs, representing approximately 18% of room nights booked, and the hotels' websites, accounting for about 17%. Interestingly, and perhaps contrary to common belief, the direct offline channel represented the most frequently used booking channel among consumers, accounting for roughly 46% of total room nights booked over our sample period.

The subject of our analysis lies at the forefront of current policy debates. Banning PPCs should enhance competition not only across sales channels but also among OTAs, potentially benefiting both end consumers and hotels. Moreover, the recently implemented Digital Markets Act (DMA) in the EU prohibits all types of PPCs for gatekeepers such as Booking.com. Drawing from our thorough analysis of a similar but smaller-scale legislation, we can identify potential challenges and offer valuable insights for policymakers and industry stakeholders.

Methodology and Results. We employ quasi-experimental methods to empirically estimate the impacts of the full prohibition of PPCs in France on hotel prices and sales shares across various channels. In our design, hotels in France are the treated group, and hotels in other EU countries, where PPCs were still allowed, serve as control units. Our analysis yields three main results.

First, we find that the ban of PPCs had a negative but not significant effect on room prices on OTAs and hotel websites, two online channels where prices are "visible" to everyone. However, a significant 5.3% price reduction was identified on the hotels' primary offline channel, which includes bookings via direct phone calls, emails, or walk-ins. Prices in this channel are usually "non-visible" to outsiders, as hotels process these transactions internally. For a typical transaction in our sample, this translates into a price reduction of about €8.5 per booking in France.

Second, our findings reveal a significant decrease in the share of room sales on OTAs, coupled with an increase in the direct offline channel. In France, bookings via the OTA channel experienced a relative decline of 2.1% compared to the controls, whereas bookings made through the main offline channel increased by about 4.6%. No statistically significant change was detected in the share of reservations made through the hotels' websites.

Third, for one of the hotel groups that provided detailed data on commission rates, we find a general reduction in average OTA fees in both 2016 and 2017, which was, however, more pronounced in France. Interestingly, the difference in percentage fee reductions between France and the control group is similar to the percentage price reductions on OTAs.

Several factors may explain these findings. On visible channels, hotels seem to refrain from noticeable price differentiation to avoid potential retaliation from platforms. Hunold *et al.* (2020) demonstrate that OTAs tend to downlist hotels that set lower prices elsewhere, a practice known as "dimming". Peitz (2022) suggests that platforms may adjust their recommendation algorithms to favour hotels with higher conversion rates, as lower conversion rates could signal that hotels offer more attractive prices on other channels. Bar-Isaac and Shelegia (2025) demonstrate that sellers' hesitation to differentiate prices may be linked to the need to maintain a high position in future ranking. Another response by OTAs to the ban of PPCs was the introduction of Preferred Partner Programmes (PPPs), where maintaining price parity became the condition for top placement in search results (Cazaubiel *et al.*, 2021; Tirole, 2023). Either way, OTAs may achieve outcomes similar to PPCs on online channels without formally applying them.

However, since monitoring prices offline is more difficult, hotels in France were able to offer lower rates through their direct non-visible channel. This pattern did not arise in other EU countries, even though major OTAs such as Booking.com and Expedia had switched from wide to narrow PPCs shortly before the Macron Law. In principle, this change could have allowed hotels to lower prices in the offline channel. A possible interpretation of this difference is that consumers in France were more aware of the opportunity to exert additional effort to find lower offline prices, potentially as a result of the broader media coverage of the Macron Law. Consequently, in relative terms, a larger share of bookings shifted offline following the PPC ban, suggesting that hotels engaged in price discrimination.[1] Nevertheless, we do not observe a significant post-policy expansion in total room-night reservations for the French hotels in our sample compared to the control ones.

Given the identified changes in both the prices and shares of the offline sales channel, we also estimate the consumer welfare implications of the Macron Law for the population of 3-star to 5-star French hotels within our sampling window. Inspired by recently developed techniques to calculate welfare bounds across consumer groups (Canzian *et al.*, 2024; Kang and Vasserman, 2025), we find non-negligible savings. Over the sampling period of our study, we estimate that consumers who booked directly through the primary offline channel that we consider may have saved up to approximately €217 million. However, this amount is relatively small, from an economic perspective, compared to the annual revenue of the entire French hotel sector, which ranged from €24 to €26 billion during the same period (Euromonitor International, 2016).

Our findings are relevant for at least two reasons. First, they suggest that hotel clientele may be segmented, with some consumers willing to exert extra effort to secure better prices by contacting hotels directly. Second, the main pro-competitive effect of the policy reform appears not on OTAs or hotel websites, as much of the economic and policy literature has suggested (see, among others, Edelman and Wright, 2015; Johnson, 2017; Baker

[1] We use the term *price differentiation* when channels entail different costs (e.g., OTAs vs. hotel website), whereas *price discrimination* applies when price differences reflect consumer sensitivity across channels with similar costs (e.g., hotel website vs. direct offline channel).

and Scott Morton, 2018), but rather on the primary offline channel where consumers can directly contact the hotels. While less visible, this channel accounts for a substantial share of transactions, as noted earlier, and we observe significant price reductions along with increases in its relative sales share. Notably, without access to proprietary data, such effects might remain undetected.

Finally, we would like to highlight the policy implications of our results. Dominant OTAs seem to have found ways to convince hotels to respect price parity for the online prices, even after the legislative ban on PPCs. In this regard, the policy may have been ineffective in achieving its intended objectives, as also suggested by the economically limited implications that we estimated. This challenge extends beyond the lodging sector, as illustrated by Amazon's practice of removing the "Buy Box" option for those products with lower prices offered elsewhere (Hunold *et al.*, 2022; Scott Morton, 2023). In this context, we suggest that additional provisions should be incorporated in regulations aiming at countering the dominant position of large platforms to make the entire market more transparent and competitive.

Institutional Context. The past decade has been characterised by a series of policy interventions against PPCs, especially when they were adopted by dominant platforms. In 2013, Amazon was forced to remove PPCs in the EU following antitrust investigations in Germany and the UK, then in the US in 2019 due to mounting political pressure. In November 2020, the UK Competition and Markets Authority (CMA) issued an unprecedented fine of almost £18 million against a price comparison website for insurance due to its use of wide PPCs.

In the lodging sector, Booking.com and other major OTAs switched from wide to narrow PPCs in the EU in 2015, and in Australia and New Zealand in 2016, following investigations by competition authorities. However, narrow PPCs raise similar concerns to wide ones, as hotels might maintain price parity to avoid diverting demand from their own commission-free websites to rival OTAs that, although cheaper than the dominant ones, still charge fees.[2] For this reason, some EU countries also prohibited narrow PPCs, starting with France in 2015, and continuing with Germany and Austria in 2016, Italy in 2017, Belgium in 2018, and Switzerland in 2022.

PPCs remain a central issue of interest for policymakers dealing with the challenges posed by dominant digital platforms. In May 2021, the German Federal Court of Justice ruled that narrow PPCs violated antitrust laws, thus confirming the decision of the Federal Cartel Office in 2015 to prohibit all types of PPCs. In November 2021, the UK CMA recommended that wide PPCs be included in the list of hardcore restrictions in the revision of the Vertical Agreements Block Exemption regulation (Marshall *et al.*, 2021). In July 2024, the Spanish Competition Authority fined Booking.com €413.24 million for abusing its dominant position in Spain between 2019 and 2024. The decision followed complaints from hotel associations about a combination of practices, including narrow PPCs and the platform's ability to unilaterally lower hotel prices.

The DMA, adopted by the European Commission in September 2022, includes the prohibition of all types of PPCs whenever gatekeeper platforms are involved, with enforcement starting in March 2024.[3] This prohibition is complemented by provisions aimed at eliminating practices that serve as substitutes for PPCs, such as dimming and other forms of

[2] The report by European Competition Network (2017) confirms that the incentive to price differentiate under narrow PPCs is very limited, as hotels see "no reason to treat [their] OTA partners differently", and do not want "their website to appear as more expensive than the OTAs" (see page 11 of European Competition Network, 2017).

[3] See Article 5.3 of the DMA, https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R1925.

algorithmic manipulations, echoing our policy recommendations.[4] Booking.com's designation as a gatekeeper in May 2024 makes the analysis in this paper particularly timely and relevant.

Structure. The remainder of the article is as follows. Section 2 reviews the related literature. Section 3 presents a simple model to derive predictions about banning PPCs on visible and non-visible channels. Section 4 presents the data and summary statistics. Section 5 outlines the empirical strategy. Section 6 reports the main results on prices, booking shares, and OTA commissions. Section 7 examines the robustness of our findings, while Section 8 explores heterogeneous effects. Section 9 discusses consumer welfare implications, and Section 10 concludes.

2 Related Literature and Contribution

Theoretical papers agree that removing all types of PPCs should lower prices both in direct channels and on OTAs by intensifying inter-channel competition and reducing commission rates (Edelman and Wright, 2015; Boik and Corts, 2016; Johnson, 2017; Wang and Wright, 2020). Following the contractual change, sellers may indeed renegotiate their agreements with platforms, paving the way for price adjustments. In this respect, chain affiliation, typically linked to stronger managerial organisation (Kosová and Lafontaine, 2012; Hollenbeck, 2017), is expected to confer greater bargaining power in negotiating commission rates with platforms.

[4]See, for example, Article 6.5, which mandates transparent, fair, and non-discriminatory ranking conditions; https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R1925.

Theoretical findings on narrow PPCs are less clear-cut, as these clauses may represent a compromise between increasing competition in the sector and rewarding OTAs for the service they provide (Edelman and Wright, 2015). Moreover, when competition across platforms is already in place, narrow PPCs may be necessary for multiple OTAs to coexist, especially if showrooming is a concern (Wang and Wright, 2020).[5] Conversely, Johansen and Vergé (2017) argue that competition remains limited with narrow PPCs, as significant commission cuts would be needed to induce price reductions, which would prove unprofitable for platforms.

Recent empirical contributions do not provide conclusive evidence either, possibly because such studies rely on heterogeneous sample compositions (e.g., chains vs. all hotels), data sources (e.g., transaction vs. offered prices), and methodological designs. On the one hand, Hunold *et al.* (2018) and Ennis *et al.* (2023) show that the (partial or full) removal of PPCs increases the likelihood that direct channels feature the lowest price. The former article compares trends in different countries following Germany's ban of all types of PPCs for Booking.com in 2015. The latter, using data from the EU and worldwide in 2014 and 2016, studies the main events of 2015, namely, the switch from wide to narrow PPCs in the EU and the full removal of PPCs in France and Germany.

On the other hand, a report commissioned by the EU in 2016 (European Competition Network, 2017) found scarce evidence of price differentiation across sales channels after the policy interventions in 2015. Moreover, Mantovani *et al.* (2021) examined the effect of the full removal of PPCs in France in 2015 using data scraped from Booking.com's website covering the period 2014–2017 for holiday destinations in France and Italy. They found a significant reduction in the short run for hotel prices on Booking.com for the French destinations,

[5] Wals and Schinkel (2018) add to Wang and Wright (2020) the possibility that platforms adopt a best price guarantee (BPG), and show that BPG strategies, when combined with narrow price parity clauses (PPCs), can serve as a substitute for wide PPCs in deterring the entry of a less efficient potential entrant.

followed by a limited response in the medium run. They also find that chain hotels exhibited a more pronounced price reaction, although weakly significant only in the medium run.

Summing up both theoretical and empirical findings, there remains a degree of uncertainty regarding the actual effects of the policy changes introduced in different EU countries over the past years on hotel prices. It has been argued that the prohibition of PPCs may have favoured more organised units, such as chain hotels, whereas small and independent hotels may have found it more difficult to break free from the influence of dominant platforms. Our findings reveal that the difficulty in taking advantage of the policy change may be widespread, with only consumers booking directly offline benefiting from lower prices.

This article reports novel empirical estimates of the effects of removing PPCs. We focus on France, the first country to prohibit all types of PPCs in the lodging sector, and extend beyond the previously discussed articles in three ways. First, we leverage a uniquely detailed proprietary dataset with channel-level transaction information from three major global hotel groups, which enables us to analyse the effects of the policy on both online and offline channels. Second, our data allows us to measure the price and sales changes across different channels rather than the probability that a specific channel (e.g., the official website) offers the lowest price. Hence, we are the first to study the effects of the prohibition of PPCs on *both* prices and sales channel shares. Third, we exploit partial yet unique information regarding OTA rates to associate the price changes with potential rate reductions following the policy intervention. We also provide consumer welfare bounds for our results (Canzian *et al.*, 2024; Kang and Vasserman, 2025).

Overall, Mantovani *et al.* (2021) is the closest article to ours, as it exploits the same policy change. The authors gathered web-scraped prices posted on Booking.com by lodging establishments located in Sardinia and Corsica, two islands in geographic proximity and with similar characteristics. Additionally, their sample includes a larger number of lodging establishments across both islands, supporting the internal validity of their study. Our study, covering hotels in 61 cities and resort destinations across seven EU countries, is less geographically focused but offers broader insights at the EU level. Mantovani *et al.* (2021) find price decreases for rooms posted on Booking.com by hotel chains one year after the Macron Law, albeit weakly significant. Conversely, we find reductions that are not statistically significant in the transaction prices observed on the OTA channel, which includes, however, all intermediary platforms.

We emphasise that the data used in this study are proprietary and cover sales on *all of the hotels' channels*. Notably, our data shows that OTAs accounted for less than 20% of room night sales during the study period, with a similar share for the hotel website. On these visible channels, we find results partly in line with the previous literature: prices on OTAs decrease, but less than those on the hotels' websites. However, these effects are rather small and not statistically significant in our analysis. Importantly, having data from all sales channels allows us to shed light on the less monitored ones, including the direct offline channel, which accounted for almost half of total sales and has been largely overlooked in previous literature and policy discussions. Indeed, we show that the most significant *price and sales effects* occurred in the primary *offline* channel. Furthermore, we employ recently developed estimation methods that leverage machine learning techniques. This allows us to enhance our empirical estimates and demonstrate the robustness of our main findings.

Our work also adds to the recent empirical evidence on the impact of regulation on online platforms, including Facebook (Benzell and Collis, 2022), Instagram (Ershov and

Mitchell, 2025) and Amazon (Gutierrez, 2022). Recent articles provide evidence related to the EU's DMA restrictions on, for example, the ban on self-preferencing (Farronato *et al.*, 2023; Reimers and Waldfogel, 2023 and Chen and Tsai, 2024) and search engine defaults (Decarolis and Li, 2023; Decarolis *et al.*, 2025). Complementary to our reduced-form design, Lasio *et al.* (2025) develop a structural model using similar data and simulate commission rate caps and DMA-style disintermediation, highlighting how different levers shift sales and margins across channels.

In order to level the playing field between dominant platforms and sellers resorting to their services, recent contributions suggested the imposition of measures such as capping commission rates (Gomes and Mantovani, 2025; Tirole and Bisceglia, 2023; Wang and Wright, 2025) and curbing recommendation biases (de Cornière and Taylor, 2019; Teh and Wright, 2020). These additional provisions are likely to play a complementary role in the design of platform regulation to ensure that the benefits of digitisation are fairly distributed and shared among all stakeholders.

3 The Economic Effects of Removing PPCs

Our investigation aims to evaluate the effects of removing PPCs on the hotels' sales channels, including both channels that are visible to everyone, such as OTA websites or a hotel's official website, and channels visible only to transaction parties (in this case, the hotel and its clients).

To understand the mechanisms underlying the removal of PPCs, we consider a stylised model in which n hotels, denoted by i ($i = 1, \dots, n$), sell their rooms through three channels.

These channels are indexed by $j = o, w, m$, where o represents OTAs, w denotes the hotels' official website, and m refers to the direct offline channel, encompassing hotel direct bookings via emails, calls, and walk-ins. As previously discussed, prices for the first two channels are publicly visible, whereas those for the third are not. While selling through any channel requires no direct cost, transactions on o require hotels to pay OTAs a commission rate, f_o. This rate is set through bargaining between OTAs and hotels, after which hotels set prices across all channels.

The demand function for a hotel-channel pair is specified according to a demand system *à la* Singh and Vives (1984), which has been extended and used in the context of platforms by Johansen and Vergé (2017), Karle *et al.* (2020), and Calzada *et al.* (2022), inter alia. In this system, each hotel and sales channel pair represents a differentiated product, and the market is not fully covered, allowing the overall demand to either expand or contract in response to price changes. Compared to these works, our framework goes beyond the dualism between the OTA and direct online channels by also considering a channel that is not publicly observable.

In particular, we define the demand system as follows:

$$D_{ij}(p_{ij}, \mathbf{p_{-ij}}) = \alpha - \beta_0(1 + \mathbb{I}_m \tau_m)p_{ij} + \beta_1 \sum_{k=1,\ldots,n}^{l=w,o,m} p_{kl}, \tag{1}$$

where D_{ij} is the demand faced by hotel i on channel j, with p_{ij} representing the price of the room in hotel i purchased through channel j, and $\mathbf{p_{-ij}}$ the vector of prices for all other hotel-channel pairs. Parameter α is the demand intercept, while β_0 and β_1 capture the sensitivity of the demand to a hotel's own price and to the prices of other channels and competing hotels, respectively. The key parameter τ_m measures the difference in price sensitivity of consumers using channel m, which remains non-visible to outsiders, and $\mathbb{I}_m$ is an indicator function that switches on when the channel is m and PPCs are banned. This higher price sensitivity may be related, for example, to the consumers' willingness to exert the additional effort of writing an email or calling the hotel when PPCs do not apply. Additional details on the model and its analyses are provided in Appendix A.

We begin by considering a situation in which the policy change does not affect the commission rates, which are taken as given. The prohibition of PPCs increases competitive pressure across channels by allowing hotels to engage in price discrimination. In equilibrium, prices may adjust differently across channels, as PPCs create an averaging effect between consumer segments with varying price sensitivities. For example, OTA booking prices may rise relative to the case in which PPCs are enforced, whereas prices on the hotel's direct channels, both visible and non-visible, decrease. We also identify cases where the removal of PPCs leads to a decrease in prices across all three channels compared to the price with PPCs, denoted as p^*.[6] In particular, we can identify the threshold value $\hat{\tau}_m$ above which competition across channels is sufficiently intense to reduce all prices after the removal of PPCs:

$$\hat{\tau}_m = \frac{2f_o(\beta_0 + \beta_1)[\beta_1(2f_o - 7) + 2\beta_0]}{\beta_0\left[\beta_1(3 + 2f_o)(2 - f_o) + \beta_0(6 - 8f_o)\right]}. \tag{2}$$

Moreover, we can compare equilibrium prices across channels once PPCs are banned. If consumers using channel m, which remains non-visible to outsiders, are more price sensitive

[6]Indeed, PPCs typically stifle price competition across channels because all prices for a room need to be the same. A hotel selling through a lower-cost channel cannot pass on those savings to consumers in the form of lower prices without also reducing the room price on OTAs, which charge higher commissions (Scott Morton, 2023).

($\tau_m \geq 0$), the price on this channel will be lower than on the hotel website: $p^*_m \leq p^*_w$. Moreover, OTAs' commission rates, f_o, effectively raise the marginal costs for hotels, resulting in a higher equilibrium price for OTAs than on the hotel website: $p^*_w < p^*_o$. Absent any price restriction due to PPCs, we thus obtain $p^*_m \leq p^*_w < p^*_o$.

Suppose now that the removal of PPCs strengthens hotels' bargaining power *vis à vis* OTAs when setting commission rates. This may occur because price differentiation allows hotels to shift demand away from platforms to other channels.[7] In this case, the equilibrium commission rate f_o may decline, reducing the cost of using OTAs for hotels, making it more likely that this lower rate translates into a lower room price on the OTA compared to the price with PPCs.

These findings are illustrated through the numerical simulations reported in Table 1, which are based on the demand system (1) for a market with two hotels. All main parameters of the model and their values are reported in the note to the table, as well as in columns (1) and (2).

The equilibrium prices per channel can be found in columns (3) to (5), with one row for the scenario with PPCs (with p^*) and the next for the scenario without PPCs. These prices highlight the crucial role of the threshold $\hat{\tau}_m$, defined in expression (2) and shown in column (6). As an illustration, suppose that the commission rate remains unaffected by the policy change, with $f_o = 0.15$ both in the presence and in the absence of PPCs (Panel A). This yields a threshold $\hat{\tau}_m = 0.033$. When the price sensitivity of consumers on the non-visible channel is relatively low (below the threshold, e.g., $\tau_m = 0.025$), the price with PPCs lies between the prices without PPCs: $p^*_m < p^*_w < p^* < p^*_o$. If, instead, consumers on

[7] We provide evidence in this direction in Section 6.2.

this channel are more price sensitive (above the threshold, e.g., $\tau_m = 0.05$), then all prices decrease following the removal of PPCs: $p_m^* < p_w^* < p_o^* < p^*$.

Table 1. *Simulated Prices With and Without PPCs.*

	(1) f_o	(2) τ_m	(3) p_m^*	(4) p_w^*	(5) p_o^*	(6) $\hat{\tau}_m$
Panel A. Fixed f_o						
PPCs	0.15	0.025	202.0	202.0	202.0	0.033
No PPCs	0.15	0.025	191.1	195.0	203.7	0.033
PPCs	0.15	0.050	202.0	202.0	202.0	0.033
No PPCs	0.15	0.050	182.8	190.3	198.8	0.033
Panel B. Varying f_o						
PPCs	0.15	0.025	202.0	202.0	202.0	0.033
No PPCs	0.11	0.025	191.4	195.3	201.5	**0.023**
PPCs	0.15	0.050	202.0	202.0	202.0	0.033
No PPCs	0.11	0.050	183.1	190.6	196.6	**0.023**

Note: The table shows results for $n = 2$, $\alpha = 100$, $\beta_0 = 1$, $\beta_1 = 0.215$, and $\tau_m = \{0.025, 0.050\}$. Examples 1 and 2 (Panel A) assume $f_o = 0.15$, while Examples 3 and 4 (Panel B) use $f_o = 0.15$ when PPCs are present and $f_o = 0.11$ when PPCs are banned.

Also note that the threshold $\hat{\tau}_m$ increases with f_o, suggesting that a potential reduction in the commission rate after the removal of PPCs would expand the parametric region where all prices decrease. This is illustrated in Panel B for the case where $f_o = 0.11$ after the ban of PPCs, which leads to $\hat{\tau}_m = 0.023$. Here, for $\tau_m = 0.025$, all prices decrease following the removal of PPCs: $p_m^* < p_w^* < p_o^* < p^*$, whereas with $f_o = 0.15$, this outcome does not occur.

The above discussion leads us to formulate the following testable hypotheses. First:

Theoretical Prediction 1. For given OTAs' commission rates, there exist parameter regions where, at equilibrium, the price under PPCs exceeds the prices of all other sales channels once these clauses are prohibited: $p^* > \max\{p_o^*, p_w^*, p_m^*\}$. Moreover, if the OTAs' rates were to decrease at equilibrium following the removal of PPCs, this result would hold in a wider parametric region.

Moreover:

Theoretical Prediction 2. If PPCs are prohibited, the expected ranking of the equilibrium sales channel prices is $p_m^* < p_w^* < p_o^*$, provided that (i) consumers on the non-visible channel m are more price sensitive, and (ii) the demand on OTAs' channel o is less sensitive than that of the hotel website w.

4 Data

Our empirical analysis leverages transaction data, originally provided to the European Commission by a hotel category association, covering 166 individual hotels across 61 cities in 7 European countries. These hotels are affiliated with 18 chain brands belonging to three of the top 10 largest international hotel groups operating in Europe and worldwide. Our sample covers three years, from June 2014 to May 2017.[8] An observation in the data is a unique hotel-month-channel combination. Each observation features the number of bookings made, the total room nights booked through that channel, and the revenue generated.

From these data, we were able to calculate the average price per room night for each channel-month and the total room nights booked per hotel-month. Room night is a standard statistical metric in the hotel industry. At the hotel level, further information is available about the star rating, the capacity (number of hotel rooms), the review score on OTAs, and additional hotel features and amenities (e.g., whether there is an in-house restaurant, bar, or spa, etc.).

This study focuses on three channels: two online – Online Travel Agency (OTA) and Official Website (WEB) – and one offline, Direct Offline (INN), which includes bookings made

[8]Our data do not allow us to clearly identify establishment entry or exit over the sample period.

through direct phone calls, emails, and walk-ins. These channels are directly or indirectly affected by the initial imposition and subsequent prohibition of PPCs. Together, they constitute approximately 80% of all reservations made to the hotels in our sample. We calculate the sales share of each reservation channel and provide additional information about the data in Appendix B.

Table 2 presents the summary statistics for the main variables in our dataset, providing information for each individual country as well as overall. Notably, during our sample period, France (11 cities) experienced the relevant policy change. Hotels in the remaining six countries (50 cities) serve as control units. On average, the hotels in our sample have 4.0 stars and a capacity of approximately 192 rooms. Review scores are very similar, with an average of 8.4 on OTAs. Each hotel sold approximately 3883 room nights per month, with an average room price of 146.6 EUR. Additionally, note that prices in France are higher than the sample average.[9]

Finally, to provide general information regarding the hospitality sector and to study the consumer welfare implications of our findings, we also collect data on the population of French and European hotels from several sources, including France's National Institute of Statistics and Economic Studies (Insee, 2024) and the DGE (Direction Générale des Entreprises, 2017). We additionally exploit Google Trends data to analyse the level of attention raised by the policy changes (Google Trends, 2025).

[9]The table also shows that hotels in the UK are somewhat different from the rest in terms of capacity as well as occupancy rate. Despite this, we believe all countries should be included to preserve the quality of our control sample. Indeed, hotels in the UK represent a substantial share of our control group (about one-quarter of the total observations). Moreover, both the British and French markets are dominated by large hotel groups, similar to the ones analysed in our study. Finally, as we explain in the following pages, our preferred estimation method exploits the richness of information in the control sample to recover counterfactual outcomes via matrix completion.

Table 2. *Summary Statistics of Hotel Characteristics By Country*

	Star Ratings	Hotel Capacity	Review Score	Occupancy Rate	Average Price	Room Nights
France	4.1	166.2	8.3	0.60	189.4	3110
(c = 11, N = 23, n = 4438)	(0.6)	(102.6)	(0.5)	(0.16)	(67.3)	(1102)
Control	3.9	196.3	8.4	0.67	139.7	4008
(c = 50, N = 143, n = 26521)	(0.6)	(132.8)	(0.4)	(0.16)	(44.7)	(1346)
Belgium	3.6	160.3	8.3	0.62	119.4	3026
(c = 7, N = 18, n = 3171)	(0.6)	(87.6)	(0.3)	(0.15)	(24.4)	(985)
Italy	4.0	192.3	8.3	0.62	137.8	3550
(c = 11, N = 32, n = 5799)	(0.5)	(111.4)	(0.3)	(0.18)	(49.2)	(1545)
Netherlands	4.1	191.7	8.5	0.69	171.9	4070
(c = 7, N = 21, n = 4114)	(0.7)	(102.5)	(0.4)	(0.15)	(57.1)	(1204)
Portugal	4.1	161.4	8.4	0.64	117.5	3147
(c = 4, N = 16, n = 3001)	(0.7)	(53.9)	(0.4)	(0.19)	(57.8)	(1213)
Spain	3.8	170.5	8.2	0.67	109.8	3264
(c = 11, N = 22, n = 3903)	(0.7)	(95.4)	(0.3)	(0.17)	(34.8)	(1277)
United Kingdom	3.9	254.9	8.5	0.75	162.2	5806
(c = 10, N = 34, n = 6533)	(0.6)	(203.8)	(0.4)	(0.12)	(38.1)	(1489)
Overall	4.0	192.1	8.4	0.66	146.6	3883
(c = 61, N = 166, n = 30959)	(0.6)	(129.2)	(0.4)	(0.16)	(48.5)	(1315)

Note: This table reports the mean hotel characteristics of each country, as well as the overall sample. Standard deviations are reported in parentheses. France is the only country that experienced the treatment, which occurred in Month 15 (August 2015). The hotels in other countries serve as control units. The number of cities is denoted using lowercase c, number of hotels is indicated using uppercase N, and observations are denoted using lowercase n. "Star Ratings" reports each hotel's average number of star ratings. "Hotel Capacity" denotes the average number of rooms per hotel. "Review Score" reports each hotel's static average review score displayed on OTAs. "Occupancy Rate" denotes the average occupancy rate. The "Average Price" column reports the average price per room sold in each hotel. "Room Nights" indicates the average monthly room-night sales of each hotel.

5 Empirical Strategy

We exploit a major legislative change in the European hospitality sector to provide evidence on the effects of prohibiting PPCs. On 6 August 2015, France enacted the "Macron Law", thus becoming the first country in the world to ban all types of PPCs imposed by OTAs on affiliated hotels. According to competition authorities and the discussed economics literature, whose focus has been on the visible channels, eliminating PPCs should significantly lower hotel prices, especially in the direct online channel (hotel website).

In addition, our analysis, which includes a non-visible booking channel, suggests that prices may decrease on both visible and non-visible sales channels when PPCs are prohibited (Theoretical Prediction 1). Further, under certain conditions on the average price sensitivity of customers in different channels, the prohibition of PPCs may induce prices to decrease more in the non-visible direct offline channel than in the visible direct online channel and the OTA channel, respectively (Theoretical Prediction 2). As our dataset contains finalised transactional information for each booking channel per month, we explore the effects of the legislation on hotel prices and room sales across the three channels.

For our difference-in-differences (DID) identification strategy, we exploit the time dimension of our dataset by considering the outcome variables of interest *before and after* the Macron Law of August 2015. We also exploit variation in the units, with French hotels being the *treated* group, whereas hotels in the never-treated countries are the *control* group. These countries are in the same economic and geographic area, and are therefore very similar from an institutional viewpoint and subject to similar economic shocks.

Two-Way Fixed Effects. As our paper focuses on a single, non-staggered treatment event, we began our analyses with a standard two-way fixed effects (TWFE) DID design:

$$Y_{it} = \delta_i + \gamma_t + \tau^{\mathrm{TWFE}} D_{it} + \varepsilon_{it}, \tag{3}$$

where i identifies a unique hotel-channel combination, and t denotes the month. The main outcome variables that we consider are $Y_{it} \in \{\ln(p_{it}) \times 100, s_{it} \times 100\}$. The former is the natural logarithm of the average monthly price, and the latter is the percentage share of total bookings finalised through each channel. We multiply both outcome variables by 100 for an easier interpretation of the results. The variable δ_i is a hotel-channel (unit) fixed effect, and γ_t is a year-month dummy variable to account for seasonality. The term D_{it} equals 1 if unit i in month t is "treated", i.e., subject to the Macron Law. The term ε_{it} is the error term. The coefficient of interest is τ^{TWFE}, which captures the average treatment effect on the treated units (henceforth, ATT), namely, the change in the outcome variables for French hotels after the Macron Law *vis-à-vis* the counterfactual case where the legislative ban on PPCs did not occur.

Our identification relies upon four main assumptions. First, we assume that only French hotels received a major exogenous "shock" – the Macron Law. However, due to the Paris terrorist attacks on 13 November 2015, which was a persistent and negative demand shock to accommodations in Paris (Insee, 2016, Table 1), our main specifications focus on estimates that exclude hotels in the French capital. As the Macron Law and the terrorist attacks in Paris are relatively close in time, it would be challenging to isolate the price effects of one event from those of the other. Regarding the hotels in the control group, we assume that they were not exposed to any exogenous shocks.

Second, we assume that there are no anticipation effects in the pre-treatment periods. However, regulatory interventions are typically announced before they are implemented, and the Macron Law coincided with other developments related to PPCs in Europe, such as the shift from wide to narrow PPCs. In Section 7, we therefore formally address several timing-related issues and provide further robustness checks.

Third, we assume that the prohibition of PPCs in France did not affect the pricing of hotels in the control group. In other words, we rule out potential spillover effects of the Macron Law, which could have led to anticipatory behaviour in the control countries, especially those where discussions about banning PPCs were ongoing. For example, Italy and Belgium, which are part of our control group, indeed banned PPCs, but only after the end of our data period.

Fourth, in the absence of the Macron Law, the potential trend of French hotel prices and room sales (channel shares) would follow, on average, a similar trajectory to those of the control group (parallel trends). To gauge evidence for this assumption, we employ the following *event study* specification:

$$Y_{it} = \delta_i + \gamma_t + \sum_{t=-14,\ t\neq -1}^{21} \beta_t M_{it} + \varepsilon_{it}, \tag{4}$$

where δ_i is the hotel-channel fixed effect, and γ_t is the month fixed effect. The dummy variables M_{it} switch on if the Macron Law is t months away and if unit i is treated. The coefficients β_t are estimated for the "leads" and "lags" of the dynamic specification, with β_{-14} to β_{-2} regarded as the "pre-trends" and β_0 to β_{21} interpreted as the dynamic path of the ATT. The error term is ε_{it}. By convention, the coefficient of period -1 is normalised to 0.

Finally, to tackle the well-known issues of biased standard errors in DID models (Bertrand *et al.*, 2004), we follow Angrist and Pischke (2008) and Abadie *et al.* (2023) and cluster the standard errors at a higher level of aggregation, namely, the city.

Matrix Completion–Nuclear Norm (Athey *et al.*, 2021). To strengthen our analysis, we also employ the Matrix Completion–Nuclear Norm (MC-NN) estimator. MC-NN originated in forecasting tasks in Computer Science and adopts machine learning techniques to impute missing potential outcomes in panel data by solving a convex optimisation problem. Specifically, it estimates a low-rank matrix that best fits the observed outcomes while penalising complexity via the nuclear norm. The estimator learns latent patterns from pre-treatment outcomes across all units, which it then uses to predict the missing counterfactuals. The ATT is computed as the difference between the observed outcomes and the imputed counterfactuals for treated units.

Unlike traditional estimators such as TWFE, which rely on strong parametric assumptions, MC-NN is flexible and data-driven. Different from synthetic control methods (Abadie, 2021), it does not construct counterfactuals as weighted averages of control units, nor does it require exact matching of pre-treatment trajectories. Compared to matching estimators that select subsets of control units based on observable similarity (Abadie and Imbens, 2011), MC-NN leverages the entire data structure across all units and time periods to recover missing counterfactuals. These features make MC-NN particularly well-suited in our context, as hotel data exhibit high volatility and strong seasonal cycles. In our application, MC–NN smooths seasonal patterns more effectively and yields more stable estimates, leading to modest efficiency gains relative to TWFE.[10]

[10] As a robustness check, we also employ a Nearest Neighbour Matching estimator (Deryugina *et al.*, 2020).

The main model follows the form:

$$\mathbf{Y} = \mathbf{L} + \Gamma \mathbb{I}_T' + \mathbf{1}_N(\Delta)' + \varepsilon, \tag{5}$$

where $\mathbf{Y}$ denote $\mathbf{Y} \in \{\mathbf{p}, \mathbf{s}\}$, the complete outcome matrices of the outcome variables. These matrices feature both the observed outcomes and the counterfactual ones. $\mathbf{L}$ denotes the low-rank $N \times T$ matrix of counterfactuals that we estimate. $\Gamma \in \mathbb{R}^{N\times 1}$ represents the hotel-channel fixed effect, and $\Delta \in \mathbb{R}^{T\times 1}$ denotes the time fixed effect. Finally, ε is a random independent error vector. The estimation involves solving:

$$\min_{\mathbf{L},\Gamma,\Delta} \left\{ \frac{1}{|\mathcal{O}|} \|\mathbf{P}_{\mathcal{O}}(\mathbf{Y} - \mathbf{L} - \Gamma \mathbb{I}_T' + \mathbb{I}_N(\Delta)')\|_F^2 + \lambda_L \|L\|_* \right\}, \tag{6}$$

where $\mathcal{O}$ denotes the number of the outcomes, and $\mathbf{P}_{\mathcal{O}}(\cdot)$ represent a partition of such a matrix. Calculating the estimator involves matrix shrinkage and the scalar associated with the penalty term, λ, which is optimally selected through a ten-fold cross-validation process. Inference is performed via a clustered non-parametric bootstrap procedure. The estimation procedure is implemented through the *gsynth* package in R provided by Xu and Liu (2022).

6 Market Effects of Prohibiting PPCs in France: Prices and Sales

6.1 The Macron Law: Effects on Prices

The estimated TWFE coefficients are reported in Table 3, Panel A. Columns (1) to (2) report the estimated price effects of the Macron Law for the two online channels, OTA and

WEB, respectively. Recall that prices on these two channels are posted on their respective websites, making them visible to everyone, including viewers and web scrapers. The TWFE coefficient (τ^{TWFE}) indicates a -1.680% change in room prices on OTAs after the prohibition of PPCs in France. The estimated coefficient for hotel websites (WEB) shows a difference of -2.004%. Both coefficients, however, are not significantly different from zero.

Table 3. *Effects of Prohibiting PPCs: Prices*

	Dependent Variable: Log Price × 100			
	OTA (1)	WEB (2)	INN (3)	All (4)
Panel A. TWFE-DID Estimates				
τ^{TWFE}	−1.680	−2.004	−5.656***	−3.490*
	(2.477)	(2.348)	(2.074)	(1.960)
RMSE	18.881	19.630	19.618	21.789
Year-Month FE	✓	✓	✓	✓
Hotel-Channel FE	✓	✓	✓	✓
Sales Share	17.7%	17.0%	45.9%	100%
Observations	5,406	5,418	5,302	29,173
No. of Hotels	157	157	156	157
Panel B. MC-NN Estimates				
$\tau^{\text{MC-NN}}$	−1.384	−1.780	−5.343**	−3.235
	(2.628)	(2.413)	(2.170)	(1.987)
RMSE	15.276	18.346	15.959	18.580
Year-Month FE	✓	✓	✓	✓
Hotel-Channel FE	✓	✓	✓	✓
Sales Share	17.7%	17.0%	45.9%	100%
Observations	5,406	5,418	5,302	29,175
No. of Hotels	157	157	156	157

Note: The table reports the estimated price effects of prohibiting PPCs. The OTA, WEB, INN column headers indicate the coefficients estimated using subsets of the data from those channels, respectively. The last column reports the estimated coefficients using data from all sales channels. Panel A reports the TWFE results estimated using Equation (3). Panel B reports the MC-NN estimates using Equation (5). RMSE denotes the estimated root mean squared errors. Year-Month FE (γ_t) indicates the year-month fixed effects. Hotel-Channel FE (δ_i) denotes the hotel-channel fixed effects. Robust standard errors are clustered at the city level and reported in parentheses. The non-parametric bootstrap procedure is performed 1,000 times. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Column (3) shows the estimated price effects for INN, the direct offline channel. As previously discussed, the reservations for this channel are made through emails, phone calls, and walk-ins. As such, the prices for these bookings are typically not visible to parties not involved in the transactions. The TWFE coefficient indicates a difference of -5.656% for INN, which translates into approximately €8.5 per booking in France. Compared to the online channels, the offline channel experienced a much larger and statistically significant price decrease. Finally, Column (4) reports the overall effects of the Macron Law on French hotel prices, estimated using data from all sales channels. The estimated coefficient τ^{TWFE} indicates an overall price effect of -3.490%, which is, however, only weakly significant at the 10% level.

In Panel B, the estimations are performed using the MC-NN method developed by Athey *et al.* (2021). The results qualitatively confirm those obtained in Panel A. The main difference is that the impact on the prices on all channels is not significant. We also note that the estimated coefficients of both panels are quantitatively similar, vouching for the robustness of our results.

Panels A and B also report the RMSEs (root mean squared errors) of both estimators. The RMSE measures the average deviation between observed and predicted outcomes, with lower values indicating a better fit. Across all columns, MC-NN consistently achieves lower RMSEs than TWFE. These improvements led us to adopt MC-NN as our preferred estimation technique for subsequent robustness checks and heterogeneity analyses.

To offer visual evidence that the parallel pre-trends assumption is satisfied for our analyses, we present the estimated dynamic ATTs in Figure 1. These results are obtained by estimating Equation (4) using MC-NN. For each graph, the vertical axis plots the percentage price changes, whereas the horizontal axis plots the months. The vertical bar at "Aug-2015" indicates the Macron Law. Panels A to D plot the percentage price differences of French hotels versus counterfactual French hotels, for the four channels corresponding to Table 3.

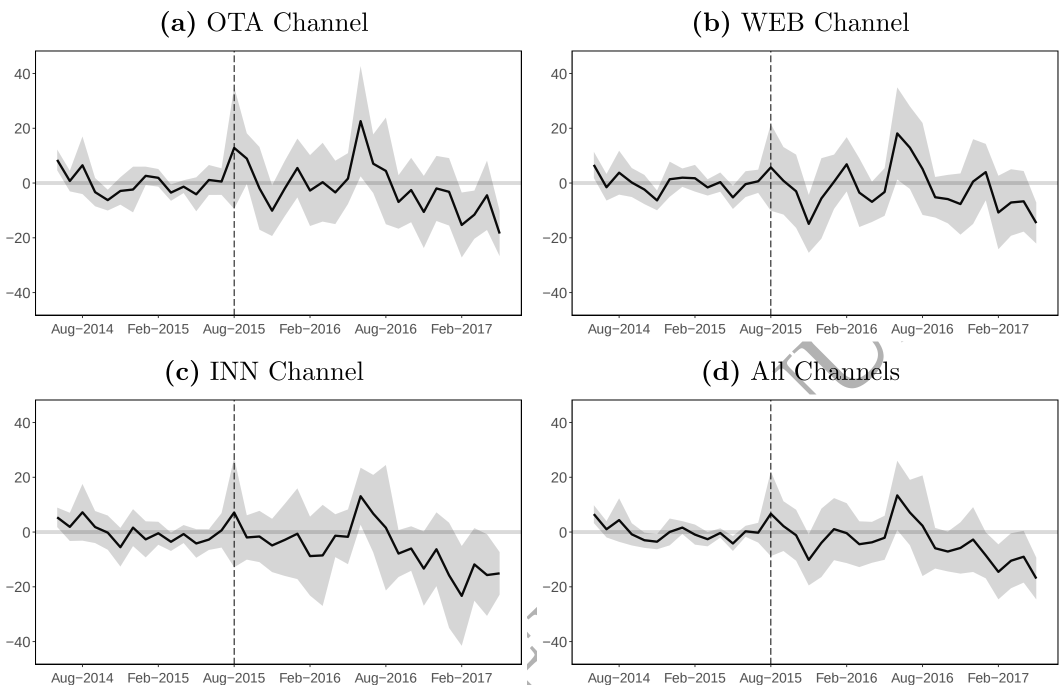

Figure 1. *MC-NN Analysis of Log Price: France and Counterfactual France.*

Examining the trends of percentage price differences before the promulgation of the Macron Law, we note that the pre-trends are relatively stable and, with rare exceptions, not significantly different from zero. This suggests satisfactory pre-treatment parallel trends and attests to the suitability of our control group and empirical strategies, particularly considering the span and diversity of our sample. The post-treatment periods confirm the mostly negative impact of the prohibition of PPCs, particularly on the INN channel. The figures also indicate a delay between the policy implementation and when the price effects materialised. We further discuss the timing of these effects in Section 6.3. We also present similar event study plots estimated using the TWFE specifications in Appendix C (see Figure C.1), which are similar to Figure 1 but more volatile due to the relative inefficiencies of the TWFE estimator.

Our first set of findings reveals an interesting picture. As noted above, most policymakers and researchers expected a significant price drop following the prohibition of all PPCs, especially on the direct online channel. In our sample, prices did decline after the Macron Law, as indicated by the negative signs in Table 3, and the decrease was greater for WEB than for OTAs. Nonetheless, neither the coefficient for OTA nor WEB is significantly different from zero, and the difference between the two coefficients is not statistically significant. Thus, Theoretical Prediction 1 is only partially confirmed. However, we observe a significant price decrease on the main non-visible offline channel (INN), supporting Theoretical Prediction 2 and suggesting that hotels in France offered better deals to customers booking through this channel after the ban.

Finding 1. The prohibition of PPCs in France led to negative but not statistically significant price effects on French hotels' visible online channels (OTA and WEB). In contrast, a larger and statistically significant negative effect (-5.343%, according to our preferred specification) was observed for prices on the non-visible direct offline channel (INN).

Our results show that significant price effects occurred in INN, the hotel's direct channel not monitored by OTAs. The other direct channel, WEB, did not experience a significant price decline. We believe that an important reason was the potential observability of such a channel by OTAs. Hence, price discrimination was adopted for the channel that was less likely to incite retaliatory responses by the platforms. In particular, practices such as the previously explained dimming (Hunold *et al.*, 2020) may have acted as a deterrent for hotels to price-differentiate on their own websites. In fact, even after the Macron Law, OTAs continued to monitor hotel pricing strategies using rate checker software and contacted them regarding any eventual parity violations (anecdotal examples are provided in Appendix D). Our interpretation is, therefore, that hotels maintained parity on the WEB channel due to the threat of OTA retaliation, but lowered prices on INN, where monitoring was weaker and consumers were more price-sensitive.

Moreover, platforms reinforced collaborative ties with hotels that respected their provisions, rewarding them with better listings and enhanced services. Booking.com, for example, visibly expanded the services offered to both customers and partner firms, especially between 2015 and 2016.[11] Interestingly, visibility boosters and partner programmes were also introduced by major OTAs starting in 2015, although we cannot determine how many hotels in our sample participated in these programmes or for how long. We further note that participation in an OTA partner programme usually guarantees enhanced visibility in exchange for higher rates and price parity across channels. However, as we will discuss in Section 6.3, OTA rates have not increased after the PPC ban for the two hotel groups that provided us with this information.

Finally, one may wonder whether hotels intensified efforts to steer consumers toward direct channels after the elimination of PPCs. Whereas we do not have direct evidence about these actions, our dataset includes partial information on outcomes such as reservations

[11]Specifically, Booking.com offered travellers tools to increase their interaction with hotels, such as the *Booking Messages Interface*, a chat tool released in May 2016 to better connect hotels and travellers, and the *Booking Experiences Tool*, introduced in July 2016, which allows users to browse a full list of activities and book tickets in advance. Another important online feature was added in 2016, as indicated by Mantovani *et al.* (2024): a hotel room was not only indicated as discounted, but the website started to include both the full price, crossed out in red, and the discounted price with the percentage reduction offered to hotel guests. While offering no real benefit, this design could enhance perceived service quality and boost hotel visibility on the platform.

by loyalty members and bookings with breakfast included, features often used by hotels as promotional tools. An increase in these proxies may indicate that hotels have tried to include these as benefits to attract users to their online and offline channels. We acknowledge that this may not be the case and that any eventual increase in the outcome may instead be driven by unobserved factors.

While we observe no significant change on the INN channel in France, both levers increased notably on the WEB channel (see Table E.1 in Appendix E). This finding is consistent with hotels concentrating non-price promotions on WEB, where OTA monitoring constrained price cuts, while offering price discounts on INN to directly target price-sensitive consumers.

6.2 The Macron Law: Effects on Booking Channel Shares and Room Night Sales

Effects on Shares of Booking Channels. We now examine the effects of the Macron Law on the sales share of each distribution channel. The sales share is calculated as the room nights sold through a given channel divided by the total room nights sold by the hotel in a given month. Compared to alternative variables (e.g., room nights), the channel share is presented as a percentage and is therefore already normalised across hotels and countries. In addition, it is directly proportional to the number of room nights sold through a specific sales channel.

Studying the changes in channel shares allows us to examine whether the hotels in our sample experienced shifts in sales across different channels. Panels A and B in Table 4 report the estimated coefficients for the two chosen estimation methods. Columns (1) to (2) report the estimated effects of the Macron Law on channel shares for OTA and WEB. The TWFE coefficient (τ^{TWFE}) indicates a statistically significant -2.121% change in the share of sales finalised through OTAs after the Macron Law. On the other hand, the estimated coefficient for hotel websites (WEB) is of lower magnitude, -0.864%, and not statistically different from zero.

Table 4. *Effects of Prohibiting PPCs: Channel Shares*

	Dependent Variable: Channel Share × 100		
	OTA (1)	WEB (2)	INN (3)
Panel A. TWFE-DID Estimates			
τ^{TWFE}	−2.121** (0.824)	−0.864 (0.702)	4.526*** (1.091)
RMSE	7.288	5.840	11.078
Year-Month FE	✓	✓	✓
Hotel-Channel FE	✓	✓	✓
Sales Share	17.7%	17.0%	45.9%
Observations	5,406	5,418	5,302
No. of Hotels	157	157	156
Panel B. MC-NN Estimates			
$\tau^{\text{MC-NN}}$	−2.149*** (0.813)	−0.889 (0.737)	4.600*** (1.039)
RMSE	7.070	5.965	10.858
Year-Month FE	✓	✓	✓
Hotel-Channel FE	✓	✓	✓
Sales Share	17.7%	17.0%	45.9%
Observations	5,406	5,418	5,302
No. of Hotels	157	157	156

Note: This table reports the estimated effects of prohibiting PPCs on channel shares. The OTA, WEB, INN column headers indicate the coefficients estimated using subsets of the data from those channels, respectively. Panel A reports the TWFE results estimated using Equation (3). Panel B reports the MC-NN estimates using Equation (5). RMSE denotes the estimated root mean squared errors. Year-Month FE (γ_t) indicates the year-month fixed effects. Hotel-Channel FE (δ_i) denotes the hotel-channel fixed effects. Robust standard errors are clustered at the city level and reported in parentheses. The non-parametric bootstrap procedure is performed 1,000 times. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interestingly, whereas we did not find statistically significant price decreases on either of these online channels (see Section 6.1), there was a relative decrease in the number of transactions finalised through OTAs. Analysing the price effects of the legislation in isolation would overlook changes in sales shares and, as a result, the overall effects of the policy may be misestimated.

Column (3) reports the estimated effects on the sales shares of the non-visible INN channel. Compared to the visible online channels, information regarding the prices or quantities (such as the number of available rooms) of the direct offline channel cannot be directly observed or retrieved by web scraping. Nonetheless, this channel constitutes a substantial share of the total room nights sold by hotels, averaging above 40%.[12]

The TWFE coefficient for INN indicates a significant increase of 4.526% after the Macron Law relative to the control. This substantial increase in the sales share of the main offline channel contrasts with the significant decrease in the OTA channel, and suggests a shift in the rooms sold from OTA to INN. Also, in this case, the estimated coefficients are quantitatively similar across Panels A and B, vouching for the robustness of our results.

Finally, Figure 2 presents the estimated dynamic ATTs.[13] The interpretation of each graph is similar to the graphs in Figure 1, with the distinction that sales share differences, rather than percentage price differences, are presented on the vertical axis. These figures offer visual evidence that the parallel pre-trends assumption is also satisfied for our sales share analyses. The post-Macron Law patterns graphically illustrate the results discussed above: a significant shift in sales occurred from the OTA channel to INN. Also, note that

[12] See Appendix B for further information on the sales shares of each channel, in particular Table B.1.
[13] Similar event study plots estimated using the TWFE specifications can be found in Appendix C (Figure C.2).

this shift took a few months to materialise, with no change in the first few months after the prohibition of PPCs in France.

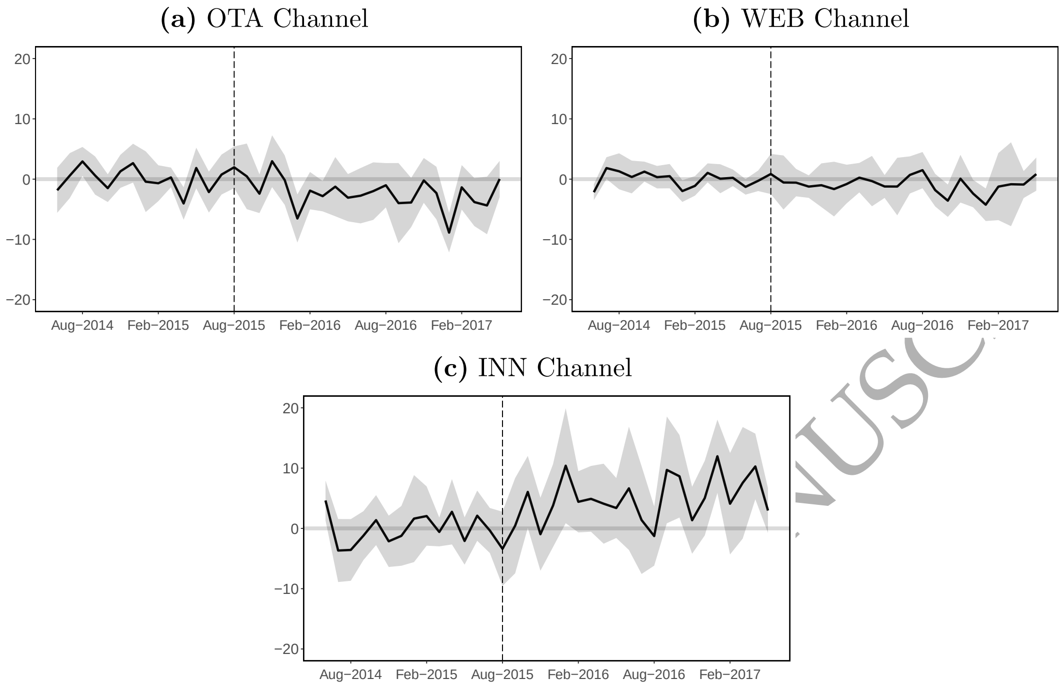

Figure 2. *MC-NN Analysis of Log Channel Share: France and Counterfactual France.*

Finding 2. The prohibition of PPCs in France led to a significant decrease in OTA sales share for French hotels (-2.149%) and a significant increase in INN sales share (+4.600%). In contrast, the share of sales through WEB showed no significant change.

Overall, the following picture emerges. When examining the price and sales effects holistically, we observe a significant sales shift from OTA to INN, but no significant change to WEB. As shown at the end of Section 6.1, hotels focused on attracting consumers to their websites by offering promotional deals. However, the findings in this section suggest that these efforts were largely ineffective. Indeed, consumers appear to respond primarily to price cuts, as evidenced by a significant increase in the sales share of the offline channel, where prices decreased.

One may wonder why similar effects were not observed in control countries, given that major OTAs had implemented similar commitments in 2015 (e.g., narrow PPCs) at the EU level, which exempted offline channels from parity obligations. It is worth observing that, while the Macron Law had a significant media impact in France, possibly raising consumer awareness and encouraging them to make an effort to book offline, no such effect was observed in the other EU countries, where the switch from wide to narrow PPCs seems to have attracted far less attention. We support this conjecture with evidence from Google Trends (see Appendix F), which indicates that the Macron Law generated substantial interest in both web and news searches in France and beyond. By contrast, the commitments of major platforms such as Booking.com and Expedia to adopt narrow PPCs received relatively little attention. However, we acknowledge that the observed offline price reductions could also be consistent with other supply-side and institutional mechanisms that do not rely on consumer awareness.

Effects on Room Night Sales and Reservations. To complement our analysis of the effects on channel shares, we briefly examine the effects of the Macron Law on room nights booked and the number of reservations made. As noted above, these two measures are highly correlated with channel shares, as the latter is simply derived by dividing the room nights sold through each channel by the total room nights sold by a hotel in a given month.

Table 5 presents the results of the MC-NN analysis, where room night sales and the number of reservations are used as dependent variables. The event study plots can be found in Appendix G (see Section G.2, Figures G.2 and G.3).[14] Similar to the results of channel shares in Table 4, we notice mild reductions in OTA and WEB sales compared to the control group, but an increase in sales through INN, although the changes are mostly weakly or not statistically significant.

Table 5. *Effects of the Macron Law on Room Nights and Reservations*

	OTA (1)	WEB (2)	INN (3)	All (4)
	Dependent Variable: Room Night Sales			
$\tau^{\text{MC-NN}}$	−44.11	−36.12*	117.4*	−6.514
	(30.91)	(21.55)	(61.47)	(17.07)
Year-Month FE	✓	✓	✓	✓
Hotel-Channel FE	✓	✓	✓	✓
Sales Share	17.7%	17.0%	45.9%	100%
Observations	5,406	5,418	5,302	29,175
No. of Hotels	157	157	156	157
	Dependent Variable: Number of Reservations			
$\tau^{\text{MC-NN}}$	−20.58	−20.16**	103.4***	5.395
	(15.61)	(9.003)	(40.15)	(9.214)
Year-Month FE	✓	✓	✓	✓
Hotel-Channel FE	✓	✓	✓	✓
Sales Share	17.7%	17.0%	45.9%	100%
Observations	5,406	5,418	5,302	29,175
No. of Hotels	157	157	156	157

Note: This table reports the estimated effects of the Macron Law on showrooming. Year-Month FE (γ_t) indicates the year-month fixed effects. Hotel-Channel FE (δ_i) denotes the hotel-channel fixed effects. Robust standard errors are clustered at the city level and reported in parentheses. The non-parametric bootstrap procedure is performed 1,000 times. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

These results suggest a shift from online sales to direct offline sales, supporting our previous analyses on channel shares. In addition, these identified patterns could potentially be consistent with showrooming. However, as we lack direct evidence on consumers' search

[14]In Section G.1, we also provide descriptive evidence that the hotels in our estimation sample did not experience significant expansions in their sales in the window of observation. The total number of room nights sold remains constant over the years, albeit with seasonal fluctuations.

behaviour, we cannot take a definitive stance on this interpretation. Other mechanisms may also align with our findings. For instance, an increased price sensitivity induced by the ban of PPCs may have led a subset of consumers to take advantage of the lower prices offered through INN, without necessarily having initially searched on OTAs.

6.3 Commission Rate Reductions and Price Effects

In this section, we examine the relationship between prices and OTAs' commission rates proposed in Section 3. The data are on the *effective commission rate* (fees paid relative to the value of OTA bookings for each month), which incorporates the contracted baseline percentage plus any adjustments linked to visibility or partner programmes where applicable. We do so by focusing on Hotel Group 1, which provided us with granular data on the OTAs' commission rates for each affiliated establishment.[15]

Figure 3 plots the average normalised OTAs' commission rates for establishments belonging to Hotel Group 1 in France and the control group, respectively. It can be observed that, several months after the Macron Law, there was an approximate 4% reduction in the OTA commission rates faced by the hotels in this group. This change occurred at the start of 2016, which most likely resulted from renegotiations between the hotel group and the OTAs. A similar but more moderate reduction can be observed in 2017, closer to the end of our sample period.

This common decrease of the OTA commission rates can be related to a number of factors. For example, it may indicate a spillover effect of the Macron Law to other countries,

[15] Hotel Group 2 also provided information about commission rates, but at a more aggregate level. This evidence can be found in Appendix H, Figure H.1.

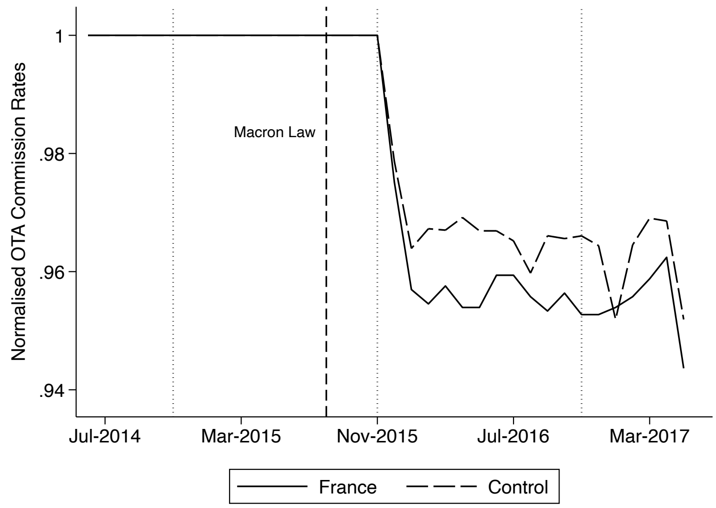

Figure 3. *Normalised OTA Commission Rates: Group 1*

but could also be the result of a downward trend in the industry.[16] It is also worth noting that the commission rate change is not exactly identical for the hotels in the treated and the control group: indeed, the French hotels enjoyed a 1.013% further reduction in 2016 (-4.444%) than their sister hotels operating in the control countries (-3.431%).

In light of the rate decrease in 2016 and the relative difference between treated and control hotels, we now perform the MC-NN analysis on hotels belonging to Hotel Group 1. The estimated coefficients are reported in Table 6, and the estimated dynamic ATT plots,

[16]During those years, hotels were trying to reduce their reliance on OTAs, for example, by trying to attract more clientele to their direct channels through "Book Direct" marketing campaigns. This may have led to a decrease in the effective rates as a result, for example, of reduced spending on ranking-boost or sponsored-placement services. At the same time, large hotel groups were aggressively bargaining to reduce their OTA rates, culminating in the well-known Marriott–Expedia deal. See: https://www.cnbc.com/2019/04/11/after-tense-negotiations-marriott-signs-a-new-multiyear-deal-with-expedia.html. More generally, we note that larger chains, like the ones in our sample, often negotiate rates and conditions with OTAs, while independent hotels have standard contracts. Unfortunately, our data do not allow us to disentangle between these two possible drivers of the decrease in the commission rates, and we acknowledge that both may have played a role.

in Appendix I. Although this exercise is only performed on a sub-sample of hotels, the pre-trends of the ATT plots are sufficiently stable, with no noticeable deviations from zero.

Table 6. *Price Effects: Hotel Group 1*

	OTA (1)	WEB (2)	INN (3)	All (4)
	Dependent Variable: Log Price × 100			
Panel A. Hotel Group 1				
$\tau^{\text{MC-NN}}$	−1.309	−2.372	−6.016**	−3.812**
	(1.941)	(2.542)	(2.431)	(1.511)
Year-Month FE	✓	✓	✓	✓
Hotel-Channel FE	✓	✓	✓	✓
Observations	2,285	2,296	2,304	12,316
No. of Hotels	64	64	64	64

Note: This table reports the estimated price effects of the Macron Law for hotels belonging to Hotel Group 1. The analyses use the MC-NN estimator. The WEB, OTA, INN column headers indicate the coefficients estimated using data subsets from the respective channels. The last column reports the estimated coefficients using data from all sales channels. Year-Month FE (γ_t) indicates the year-month fixed effects. Hotel-Channel FE (δ_i) denotes hotel-channel fixed effects. Robust standard errors are clustered at the city level and reported in parentheses. The non-parametric bootstrap procedure is performed 1,000 times. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Focusing on the effects of the PPC ban, we begin with Table 6, Column (1). The reduction in OTA prices for Hotel Group 1 (-1.309%) is similar to the full-sample estimate (-1.680%) in Table 3, Column (1). As in the baseline, the price decrease on OTA for Hotel Group 1 is not statistically different from zero. Moreover, the magnitude of the percentage price estimate is in line with the relative percentage reduction in OTA commission rates for this group, which is around -1.013%, from the data presented in Figure 3.

Finding 3. The average OTAs' rates for establishments in Hotel Group 1 decreased in 2016 and 2017. The percentage price reductions of French hotels on OTAs after the prohibition of PPCs in France are in line with the percentage reduction in the fees.

Taken together, these insights suggest that, despite the bargaining power of this internationally renowned hotel group, its establishments lowered prices only in proportion to the percentage reductions in OTA commission rates when faced with a legislative ban on PPCs. In other words, there seem to have been limited *between-channel* pro-competitive effect of removing PPCs. Any detected effect should be interpreted cautiously given the lack of statistical significance.

Table 6, Columns (2) to (4), presents the estimated price effects of the Macron Law on hotels belonging to Hotel Group 1 for the WEB, INN and All channels, respectively. The estimated coefficient for WEB is not statistically significant, which is qualitatively similar to the results of the overall analyses in Table 3, Column (2). The coefficient on INN in Column (3) is -6.016%, which is statistically significant and slightly greater in magnitude than the baseline coefficient (see Table 3, Column (3)). Finally, the price reduction across all sales channels in Column (4) is consistent in magnitude with the baseline results in Table 3, Column (4).[17]

These results on the changes in the OTA rates offer a possible rationalisation for the lagged price effects of the Macron Law identified in Section 6.1. Moreover, these findings are also in line with our theoretical predictions regarding the impact of a decrease in OTAs' rates (see the final part of Theoretical Prediction 1).

We also highlight the link between the theoretical and empirical results. The model's flexibility allows for multiple outcomes that can be mapped onto our empirical findings, with some caveats. In particular, assuming high consumer sensitivity to offline price options, the price reduction observed after the PPC ban aligns qualitatively with the model's prediction,

[17] The results in terms of the shares of each sales channel, available in Table I.1 in Appendix I, are also in line with the baseline results in Section 6.2.

in which direct prices fall below PPC prices. The example in the first two rows of Panel B in Table 1 reproduces a roughly 5% price decline in INN when PPCs are banned, consistent with Finding 1. Price reductions in the OTA channel also align with Finding 3. The only notable discrepancy concerns WEB prices, which fall less in reality than the model predicts. This is because the model does not account for OTAs' threats of retaliation. In practice, hotels kept WEB prices close to OTA levels, whereas the theoretical model, absent PPCs and without retaliation, would predict pronounced reductions of prices on the WEB channel.

7 Robustness Checks and Other Major Events

In this section, we address several relevant issues related to our identification strategy and possible threats to it, as well as further tests of the robustness of our results.

First, in Appendix J, we extend our baseline analysis by including the observations from hotels in Paris, which were subject to a severe terrorist attack in November 2015. Including these observations leaves the main findings qualitatively unaffected, but it affects the magnitude of the effects compared to Table 3. Indeed, the estimates presented in Table J.1 suggest that the negative price changes are not statistically significant on the OTA and WEB channels. However, they are larger and statistically significant on the direct offline channel, INN. All the coefficients are larger in magnitude compared to our baseline. This may raise the question of whether the decision to exclude Paris hotels from our sample is well-founded. However, our data clearly show that Paris was significantly affected by the November 2015 attacks, with substantial declines in hotel occupancy rates (Figure J.1), thereby supporting our choice of the baseline specification.

The 2016 terrorist attack in Nice may also have affected hotel demand in France, although it occurred well after the implementation of the Macron Law. In Appendix K, we perform an additional robustness check in which we re-estimate our main specification excluding observations from *both* Paris and Nice. The results are qualitatively similar to those obtained in the baseline.

Apart from these major tragic events, Insee monthly data indicate that hotel room night sales in France follow the expected seasonal pattern and, in the rest of the country, evolve along a broadly stable yearly trend (see Figure G.1, Panels C and D, in Appendix G.1). While some events, such as large sporting or cultural occasions, may cause temporary increases in local demand, our evidence does not point to any sustained structural change in hotel demand in France outside the Parisian region over the period we examine.

Second, as the Macron Law was approved by the French Parliament between June and July 2015, we also consider the possibility of anticipatory and lagged effects in Appendix L. Table L.1 presents the estimated coefficients using MC-NN when the treatment period is shifted one and two months before the Macron Law. This accounts for the OTAs and the hotels' possible anticipation of the policy. Table L.2 presents the estimated coefficients when the treatment period is shifted one and two months after the Macron Law to account for potential gaps between room reservations and check-in. The results are qualitatively and quantitatively similar to our main findings. If anything, our analysis suggests that the Macron Law was not anticipated, and its effects may be lagged, as the estimated coefficients become slightly more pronounced as we shift the treatment period later. This is consistent with our discussion in Section 6.3.

Third, during our study period, high-profile antitrust cases were ongoing in 2014 in Sweden, Italy, and France. These culminated in a commitment by Booking.com and Expedia, Europe's largest OTA platforms, to switch from wide to narrow PPCs in their contracts with hotels in the EU. This was announced in April 2015 and implemented in July 2015. This contractual change allowed sellers to lower prices not only on rival platforms but also on unmonitored channels (such as phone or in-person bookings), provided the prices were not advertised on the hotels' website. Our descriptive data (Appendix M) indicates no visible price changes following the switch from wide to narrow PPCs, with price trends continuing to follow the same seasonal patterns as before between April (announcement) and August 2015 (Macron Law).

Several factors may help explain the price unresponsiveness to the switch from wide to narrow PPCs. For offline prices, as discussed in Section 6.2, the low consumer awareness of this policy change, especially when contrasted with the wider media coverage of the Macron Law, likely played a role in the policy's lack of impact. For online prices, recall that direct bookings do not incur commission fees, whereas OTAs charge a percentage. Consequently, hotels had an incentive to maintain price parity in order to avoid diverting demand from their own website to other platforms, which, although cheaper than the dominant ones, still charged a commission. Policy reports, such as the ones by HOTREC (2015) and European Competition Network (2017), support this view and highlight the limited effectiveness of the 2015 EU reform, an outcome also consistent with theoretical predictions that narrow PPCs do little to change incentives in practice (Johansen and Vergé, 2017; Wals and Schinkel, 2018).

Fourth, we have assumed that the policy did not affect the pricing of hotels in the control group. Whereas we cannot fully rule out the possibility of spillover effects from the French PPC ban on hotel pricing in other countries, our data suggest a seasonal and generally increasing pattern in both the treated and control groups that continued after the implementation of the Macron Law (again, see Appendix M). If spillover effects did occur, though the descriptive evidence seems to suggest otherwise, our estimated price changes would have to be interpreted as a lower bound of the policy effect.

Finally, Appendix N presents two additional robustness checks. First, we provide a more direct analysis of channel price differences using a triple-difference approach. Second, we apply a nearest-neighbour matching approach after performing a seasonal adjustment to the data following Deryugina *et al.* (2020). Our main results remain robust to these additional checks.

8 Heterogeneity of the Price Effects

Heterogeneity by Pre-Treatment OTA Reliance. A moderating factor influencing price changes could be hotels' relative reliance on OTAs. HOTREC (2020) showed that OTAs' market share in the European hotel sector steadily increased from 19.7% in 2013 to 29.9% in 2019. In our sample, prior to the Macron Law, the mean sales share of OTAs was 19.2%, with a standard deviation of 11.9%, which is slightly below the European average.

On this basis, we define a hotel as having a relatively high reliance on OTAs if its pre-Macron Law OTA sales share exceeds 20%. The idea is that less-reliant hotels may be more comfortable exploiting the pricing flexibility enabled by the PPC ban, while those more dependent on OTAs might be hesitant to raise prices, fearing reduced visibility or performance on the platforms, even in the absence of legal constraints.

We then divide hotels into two categories based on their pre-treatment reliance on OTAs (Appendix O.1). Appendix O.2 provides further details on the MC-NN estimation, as well as ATT plots and estimated effects for both categories. Table O.2 presents the estimated price effects of the Macron Law: Panel A reports results for hotels with a pre-treatment OTA share below 20%, and Panel B for those above 20%. The magnitude of price reductions is consistently larger in Panel A, especially for the direct offline channel (INN) in Column (3), where we find a statistically significant decrease of -6.913%, compared to a non-significant -2.860% in Panel B.

These findings are consistent with our expectations: hotels that were *ex ante* more reliant on OTAs for their sales are also likely to be more concerned about the potential (implicit or explicit) penalties occurring if they stop respecting price parities on OTAs. Conversely, hotels that were *ex ante* less reliant on OTAs appear to be less affected by any ranking or algorithmic changes imposed by OTAs following a differential pricing strategy across different sales channels.

Heterogeneity by Pre-Treatment Occupancy Rate. Occupancy may also act as a moderating factor, as hotels with relatively high occupancy may be more resilient to shocks from any particular booking channel. Hence, they may be more inclined to experiment and differentiate prices across various channels once PPCs are no longer legally enforced. Provided that occupancy is below full capacity, this can be consistent with offering lower rates or discounts to customers on a particular channel. The occupancy in the hotels of our sample before the implementation of the Macron Law was, on average, 66.2%, with a standard deviation of 15.8%.

Appendix O.3 presents the MC-NN estimation in Table O.3. Panel A shows the estimated price effects of the Macron Law for hotels with an average pre-treatment occupancy below 65%, while Panel B reports the estimates for hotels above this threshold. Overall, price reductions are smaller in Panel A. The differences across Columns (1) to (3) are relevant, amounting to several percentage points per channel. Notably, the only significant estimate in Panel A is the −4.606% change for INN. The estimated price reductions in Panel B, however, are all statistically different from zero, ranging from −3.749% on WEB, −4.959% on OTA, to a pronounced −7.779% on INN. All of these effects are larger in magnitude than those reported in Table 3, Panel B.

Once again, these findings align with our initial expectations. Indeed, hotels with relatively high occupancy rates would be less exposed to booking channel shocks, such as a potential dimming of their search ranking on OTAs. Conversely, hotels with relatively low occupancy rates would more likely suffer from reduced visibility on online platforms.

Heterogeneity by Star Rating. We also examine the heterogeneous responses of hotels with different star ratings, which will also be relevant for the consumer welfare implications in Section 9. We note that traditional hotel segment classifications, or star ratings, tend to correlate with the quality of the hotel. Higher-star hotels typically offer higher quality and cater more to business travellers or less price-sensitive customers, whereas lower-star hotels tend to serve leisure travellers and more price-sensitive segments. As our theoretical model links the expected impact of banning PPCs with the price sensitivity of consumers, it is worth investigating whether the Macron Law had differential impacts on hotels with different star ratings.[18]

Appendix O.4 provides details on the estimation strategy, which, to preserve power, pools observations across all sales channels but subsets the data by hotel star rating. The estimated price effects of the Macron Law by star rating are reported in Table O.4, and Figure O.5 presents the event study plots. On the one hand, the estimated coefficients for 3-star and 4-star hotels are between -2% and -3%, and they are not significantly different from zero. On the other hand, the estimated coefficient for 5-star hotels is around -6.212% and is statistically significant.

These findings suggest that 3- and 4-star hotels were reluctant to lower prices after the Macron Law, possibly due to OTA retaliation. At the same time, 5-star luxury hotels show a larger coefficient than the overall price effect in the baseline model, which was -3.235% (see Table 3, Column (4)). Despite having the least price-sensitive clientele, these findings seem to indicate that luxury hotels are the main drivers of the overall negative price effect.

The intuition behind these results is that 5-star hotels, the flagship establishments of any hotel chain, tend to attract the most loyal and consistently returning clientele. As a result, these establishments are the least dependent ones (among all star ratings) on OTAs to place their capacity. Moreover, it is likely that the OTAs themselves may wish to maintain cooperation with these hotels to cultivate their presence in the high-end market. Finally, top-tier hotels often attract bookings from corporations or organisations that reserve rooms for short-term conferences or events, potentially benefiting from favourable pricing through the INN channel.

[18]We note that the hotels in our sample, on average, have relatively high star ratings (4 stars). As a result, one might expect their average consumers to be less price-sensitive, and thus the impact of the PPC ban to be weaker than it would be in the broader hotel population across all quality levels.

9 Implications for Consumer Welfare

We now try to gauge the potential magnitude of the estimated effects on consumer welfare in France by combining our sample with data on the full population of French and European hotels (2014–2017). We use external sources (DGE and Eurostat) to calibrate our sample to the characteristics of French hotels. However, the resulting welfare estimates should be viewed more as a thought experiment, as we cannot definitively claim that they generalise to all hotels in France. Moreover, these estimates capture only the pure price and quantity effects of prohibiting PPCs. Without further structure, we cannot account for potential changes in room or service quality, search costs, or consumer preferences for different sales channels.

Under these caveats, we adopt an approach inspired by Kang and Vasserman (2025) and similar to Canzian *et al.* (2024), which estimates consumer surplus using the bounds developed by Varian (1985). This method does not require any equilibrium assumptions regarding firm behaviour or optimal pricing, as would be the case in a structural approach (Canzian *et al.*, 2024). Further details on the external data and the methodological implementation are in Appendix P.

We focus our consumer welfare analysis on INN, which not only shows significant changes in prices and sales shares but also represents 46% of room nights booked in our data. The estimated relative effects for INN were also much larger in magnitude than those for online channels (Table 3). Given that the sales share of INN is approximately the same as the two online channels combined, the overall magnitude for the online channels would also likely be smaller. As a result, by considering only the INN channel, we provide a lower bound for the consumers' gains. Furthermore, since we found no significant price effects on the OTA and

WEB channels, we cannot rule out the possibility that the relative price changes induced by the Macron Law on these channels were zero.

Finally, because our sample only consists of hotels rated 3 stars or above, we limit our analysis to French hotels of the same ratings. Although this is a limitation of our approach, it is likely that the relative gains in consumer surplus for lower-rated hotels are smaller, as hotels with one or two stars may have an even higher reliance on OTAs.

To estimate consumer surplus, we leverage the availability of both price and quantity (room nights) data in our dataset, using subscripts 0 and 1 to denote the value of prices (p_0 and p_1) and room nights (q_0 and q_1) before and after the Macron Law. Following Canzian *et al.* (2024), we estimate the bounds for changes in consumer surplus (Δ CS) of the Macron Law on hotels in our sample, which only assumes that the demand is decreasing in prices:

$$\Delta \text{ CS }_{\text{Varian}} \in [(p_0 - p_1)q_1, (p_0 - p_1)q_0]. \tag{7}$$

Table 7 presents the Varian bounds for the relative changes in consumer surplus induced by the Macron Law for the direct offline channel, INN. We focus on hotel categorisation by star ratings, as this is a standard metric in the hospitality industry. Data for the population of French hotels in each category is provided by Direction Générale des Entreprises (2017). The relative gains in consumer surplus are calculated up to May 2017, the end of our sampling period. The Varian bounds are calculated by first estimating the Δ CS for an average hotel of our sample (in each star rating) using Equation (7), then multiplying by the number of hotels (in each star rating) of the entire French hotel sector.

We also account for the differences between our sample and the population by scaling the consumer welfare gains using the ratio of occupancy rates (Eurostat, 2024). The results are expressed in millions of euros. In terms of the population of French 3-star hotels, we estimated that consumers who booked directly saved up to €88.7 million in the 21 months following the implementation of the Macron Law. The corresponding savings for 4-star hotels were up to €55.1 million, and €73.0 million for 5-star hotels. The overall estimates suggest that guests of French hotels may have saved between €180 and €216.8 million in the months following the promulgation of the Macron Law, relative to their counterparts in other EU countries. To put these figures into perspective, we note, however, that the estimated overall annual revenues of the hotel sector in France were approximately €24-26 billion (Euromonitor International, 2016) during the period of our study, indicating that the economic benefits of the policy change were relatively limited.

Table 7. *Relative Gains in Consumer Surplus (France)*

	Varian Lower Bound (1)	Varian Upper Bound (2)
3-Star	70.0	88.7
4-Star	46.7	55.1
5-Star	63.3	73.0
Overall	180.0	216.8

Note: This table reports the approximated gains in consumer surplus for French hotels following the Macron Law up to June 2017. The figures are approximated using the number of French hotels in January 2017. The gains are calculated for the direct offline channel, INN, which underwent statistically significant price reductions following the Macron Law. The bounds are calculated following the procedures proposed by Varian (1985) and calculated using Equation (7). The units are millions of euros.

10 Concluding Remarks

In this article, we provided a comprehensive empirical evaluation of the impact of the Macron Law, which was introduced in France in 2015 as the first-of-its-kind legislative ban on all types of price parity clauses (PPCs) in the lodging sector. We mainly focused on the price effects of this significant policy change, while also examining its impact on the redistribution of sales shares across different channels, as well as on the OTAs' commission fees. We leveraged a unique proprietary dataset of chain hotel prices from 2014 to 2017, including all sales channels.

Our analyses, based on two estimation methodologies, TWFE DID and MC-NN, indicated that the removal of PPCs had negative but not statistically significant effects on room prices posted on OTAs and hotel websites, the visible online channels. However, we identified a significant price reduction on the hotels' main offline channel. In addition, we observed a significant decrease in the sales share of OTAs, accompanied by an increase in the shares of the direct offline channel. These results proved to be robust across various specifications and estimation techniques, including anticipatory reactions and lagged effects of the legislation.

Using granular data from one hotel group in our sample, we also found that a decrease in OTA rates was associated with a comparable, but not statistically significant, percentage reduction in online prices. By contrast, prices on the direct offline channel fell significantly for French hotels within that group.

Our findings reveal the main pro-competitive effect of the policy reform did not materialise on OTAs or hotels' websites, as initially expected, but rather on the main offline channel, where information is exclusively shared between hotels and their clients. This insight is particularly relevant, as it reflects real-world consumer diversity. Especially in digital markets, research should account for this heterogeneity and keep track of the evolving composition of consumers.

In addition, we also estimated the consumer welfare implications of the Macron Law and found substantial savings for consumers who booked directly through the offline channel, adding a quantifiable dimension to the policy's impact. The range of overall savings, relative to their counterparts in the rest of the EU, amounts to hundreds of millions of euros.

One may wonder about the external validity of our findings, in particular, whether significant price reductions occurred in other types of establishments, such as smaller chains or independent hotels. On the one hand, we believe that offering discounts on the direct offline channel to induce a shift in consumer behaviour could also be adopted by these hotels. Applying a discount to phone or email reservations is straightforward and should be feasible even for establishments with more limited managerial resources or simpler organisational structures. On the other hand, smaller chains and independent hotels may face additional challenges that large international groups are better equipped to overcome. First, chain hotels have well-established websites that are easy to find through search engines, giving them demand-side advantages (Hollenbeck, 2017) and reducing their reliance on OTAs. Second, they are generally more agile in sharing information (Baum and Ingram, 1998) and in handling the complexity of price setting (Abrate and Viglia, 2016). Third, chain hotels enjoy stronger bargaining power than independent hotels in negotiating with OTAs and resisting retaliatory strategies (De los Santos *et al.*, 2025). These factors suggest that chain hotels are well-positioned to benefit from the increased pricing flexibility and competition brought about by the PPC ban.

Despite the comprehensive dataset utilised in our analysis, certain limitations were encountered. First, the OTA channel encompasses transactions completed on major platforms, such as Expedia and Booking.com, as well as smaller ones like Ctrip or Hotel.de. Unfortunately, we were unable to distinguish between OTAs, which prevented us from analysing the differential impact of prohibiting PPCs on specific platforms. Yet, it should be noted that, at the time of our analysis, the EU OTA sector was characterised by high concentration: according to HOTREC (2018), as of 2017, 65.6% of all OTA transactions took place on Booking.com. Together with Expedia and HRS, the aggregate market share reached about 86%.

In addition, the existing literature suggests that the removal of PPCs may lead to the entry of new OTAs (Ezrachi, 2015) but could also stifle incumbent platforms' propensity to invest and innovate (Wang and Wright, 2022). Regrettably, issues related to market entry and innovative activities carried out by OTAs fall outside the scope of this paper. Moreover, although we extrapolated our empirical findings to the population of 3-star to 5-star hotels in France, we were unable to do so for the unrated, 1-star, and 2-star hotels due to limited quantitative information on how the Macron Law would impact them.

Notwithstanding these limitations, our evidence shows that eliminating PPCs alone did not lead to substantial price reductions in the intended channels. Significant effects emerged only on hotels' direct offline channels, which were not monitored by OTAs, suggesting that practices like dimming may have discouraged price differentiation on hotel websites and OTAs. Moreover, for the channel in which prices significantly declined, our estimates indicate that consumer benefits were modest relative to the overall size of the industry.

Our results suggest that policymakers and antitrust authorities should anticipate how dominant platforms might respond to policy changes aimed at curbing anti-competitive practices. Failing to do so may result in a plethora of interventions that require consistent resources and efforts, without necessarily achieving their policy goals. As discussed in the Introduction, the DMA not only explicitly bans PPCs for gatekeepers but also targets substitute practices, such as algorithmic manipulations, offering a broader set of provisions likely to be more effective than a simple PPC ban in promoting competition across sales channels. However, recent cases, such as the EU non-compliance investigations into Apple and Meta under the DMA, highlight the need for ongoing regulatory and monitoring efforts to safeguard competition in digital markets.

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Supplementary data

The data and codes for this paper are available on the Journal repository. They were checked for their ability to reproduce the results presented in the paper. The authors were granted an exemption to publish parts of their data because access to these data is restricted. However, the authors provided the Journal with temporary access to the data, which enabled the Journal to run their codes. The codes for the parts subject to exemption are also available on the Journal repository. The restricted access data and these codes were also checked for their ability to reproduce the results presented in the paper. The replication package for this paper is available at the following address: https://doi.org/10.5281/zenodo.19661103.

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